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A PALEOMAGNETIC STUDY OF VOLCANIC ROCKS FROM MISTASTIN LAKE, LABRADOR, CANADA

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Paleomagnetic data are presented for an isolated occurrence of unaltered and undisturbed volcanic rocks, of apparent upper Triassic age, at Mistastin Lake, western Labrador. Pilot tests established that the magnetization of these rocks is remarkably stable and homogeneous, suggesting that it was acquired by the rocks at the time of their original cooling. After a.f. cleaning in a peak field of 250 Oe, 73 cores (drilled at 10 sites distributed over an area of about 100 km^2), yielded a mean direction of magnetization: declination 179.9°, inclination −68.2°. A statistical analysis of the data indicates that mean directions of magnetization of the 10 sites are significantly different. Their angular standard deviation is 4.4°, and the radius of the circle of confidence about their resultant ($\alpha'_{0.95}$) is 2.4°. The corresponding virtual pole position is 117.7°E, 85.5°N (computed from the resultant of the 'weighted' site means). The good agreement between this pole and the only other published, reliable Jurassic pole from North American rocks suggests (1) that the magnetic field was mainly dipolar during part of upper Triassic to Lower Jurassic time, and (2) the radiometric age of the Mistastin Lake volcanic rocks (202 m.y.) may be slightly too old. The low standard deviation of the site means is concordant with field observation that the eruptions of the various phases of the Mistastin Lake volcanics were almost simultaneous, and that the rocks have been almost undisturbed since original deposition.

1. INTRODUCTION

Mistastin Lake (63°22'W, 55°52'N) occupies an elliptical northeast trending depression, approximately 17 by 11 km in size, cut into the moderately rugged, barren hills of western Labrador (fig. 1). The lake is surrounded by small buttes of flat lying volcanic rocks. A sample from a butte at the west end of the lake gave a whole rock potassium-argon age of 202 ± 13 million years [1]. Metamorphosed Precambrian rocks, cut by dikes of igneous rocks, are exposed on islands in the lake [2]. The structure as a whole is a crater of the type of the Clearwater Lakes [3], and Manicouagan [4].

This isolated occurrence of presumably Upper Triassic rocks offers a favourable opportunity to (1) complement our scanty paleomagnetic data for this period, (2) determine the minimum angular standard deviation of paleomagnetic vectors obtainable over an area of about 100 km^2 under optimum conditions, (3) verify their assumed Jurassic age (pending further radiometric age determinations), by comparing their paleomagnetic pole position with that obtained from a well dated suite of plutonic rocks in the White Mountains of New Hampshire (some 15° of latitude south of their own locality), (4) assess the dipolar nature of the paleomagnetic field during Jurassic time by comparing these two poles. This paper describes and analyzes paleomagnetic data obtained from the volcanic rocks of Mistastin Lake.

2. GEOLOGICAL SETTING

A geological sketch of the structure, and of the sample localities, is shown in fig. 1. Precambrian rocks in this area consist of generally granitoid rocks, commonly with spectacular augen texture, containing

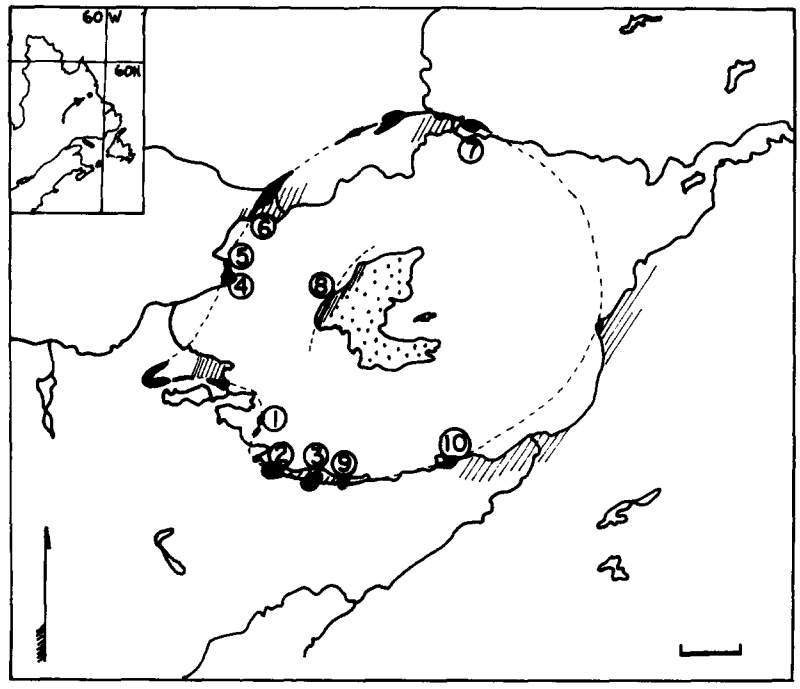


Fig. 1. Generalized geological map showing sampling localities. Sample sites numbered and circled. Blank - Precambrian, black - volcanics, hatchured - brecciated country rocks, dotted - metamorphosed Precambrian.

curvilinear belts, a few miles in extent, of white to blue anorthosite and green mangerite.

Volcanic rocks outcrop in a narrow arcuate belt which has been traced for 40 km around the northern, western and southern sides of the lake. To the east, the presumed continuation of the belt is covered by the lake. Where the volcanic belt is dissected by canyons, the lowest part is found to consist of a composite ring dike about 30 m wide which is believed to follow a ring fault in the Precambrian rocks. The margins of the dike comprise a few centimeters of flow-banded, intensely vesicular volcanic breccia, grading abruptly inward to dense, black, aphanitic andesite. A narrow, dike-like core of intensely vesicular black andesite is usually present. A small part of a similar, probably concentric, dike outcrops on the northwest shore of the central island.

Lobate or elliptical areas of flat lying volcanic rock up to 800 m in diameter are fed by this composite dike. The basal unit in each of these flat-lying sequences is a vesicular, brick red, volcanic breccia with aphanitic matrix (sample unit 1). Vesicular black andesite overlies this breccia, either gradationally or with slight angular unconformity (unit 2). Dense black andesite (unit 3) caps all except two occurrences. It grades downward to vesicular andesite, and is cut by dikes of similar material. Remnants of a sheet of pale grey pumice-breccia similar to suevite overlie the andesite at two localities. The fragments in this breccia appear to be almost exclusively of other volcanic rocks. The maximum measured thickness of volcanic rocks is about 40 m.

The field evidence suggests that all of the volcanic rocks are of very similar age. The top and bottom volcanic breccia units may be of slightly different age, but probably a single series of eruptions from several points on the ring dike produced all of the sampled rocks. The horizontal character of the sheets of volcanic rocks, and the apparent lack of faults in them suggest that no major structural disturbance has occurred in this area since their emplacement.

Microscopically, the andesites are very fine-grained aggregates of microlitic plagioclase, which may have rosette or dendritic texture, and an aphanitic, almost opaque, matrix which contains abundant, very fine-grained magnetite. Somewhat

larger octahedra of magnetite occur sparingly, apparently as phenocrysts. The red volcanic breccia contains hematite as the opaque mineral evenly distributed throughout the aphanitic matrix in minute globules. The inclusions are partially digested, and completely recrystallized. The rocks are remarkably fresh. No evidence of any alteration at all was found either in the field or in thin section.

3. SAMPLING AND MAGNETIZATION MEASUREMENTS

A total of 76 cores from ten sites, covering the geographic and petrographic range of the volcanic rocks, were drilled with a portable diamond drill. At each site, cores 3.18 cm in diameter were drilled about 3 m apart and oriented *in situ* with a solar compass when possible, or with a Brunton compass when the sun was invisible. At least half the cores at each site were oriented with both solar compass and Brunton compass, and the local magnetic declination derived from these pairs of observations was used to correct the orientation of cores oriented only with the Brunton compass.

The magnetization measurements reported below were performed on a three-magnet, automatic, astatic magnetometer which measures magnetization intensities with an accuracy of 1 percent and magnetization directions repeatable within 0.5° [5].

4. MAGNETIC STABILITY TESTS

One specimen from each of the ten sites was submitted to a series of alternating-field 'cleaning' tests on the apparatus of Larochelle and Black [6]. The peak value of the demagnetizing alternating field was increased by successive steps of 50 or 100 Oe up to 500 Oe. The intensity and direction of the magnetization determined after each treatment are plotted in figs. 2 and 3 respectively.

As illustrated in these diagrams, the rocks appear to have a remarkably high magnetic stability. The minor component of viscous magnetization in some of the untreated specimens completely vanished after the first alternating field treatment in a demagnetizing field of 100 Oe. The fact that all specimens were

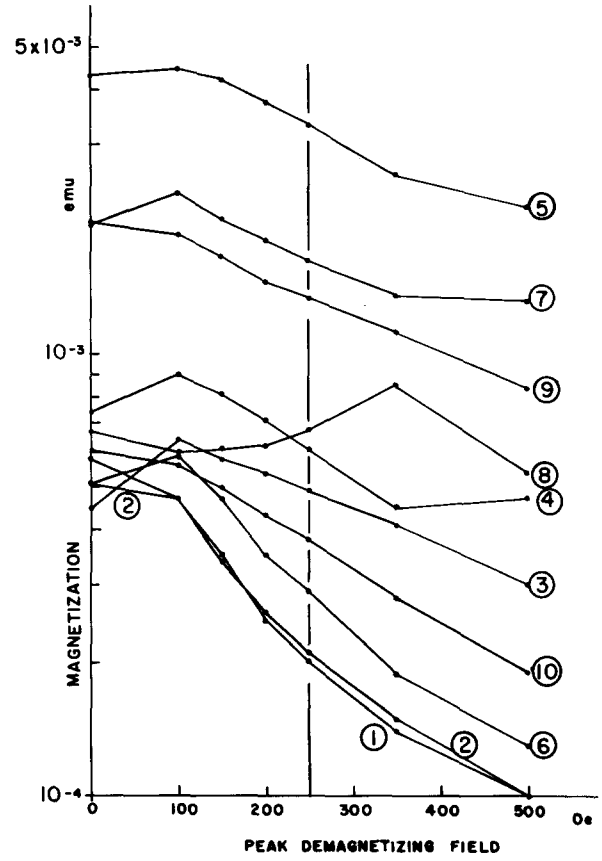


Fig. 2. Demagnetization curves of 10 pilot specimens. Circled numbers refer to site numbers of pilot specimens.

magnetized in a direction almost anti-parallel to that of the present earth's field prior to any cleaning treatment also points to the high magnetic stability of the rocks.

5. NATURAL AND STABLE REMANENT MAGNETIZATION

The intensity and direction of the natural remanent magnetization (NRM) of the remaining 66 cores were determined with the procedure explained above. For 46 of the cores the direction of magnetization was obtained by averaging the two directions obtained from two 2.81 cm long cylindrical specimens cut from each core. The direction of magnetization of the other 30 cores was based on the measurement of one specimen for each core.

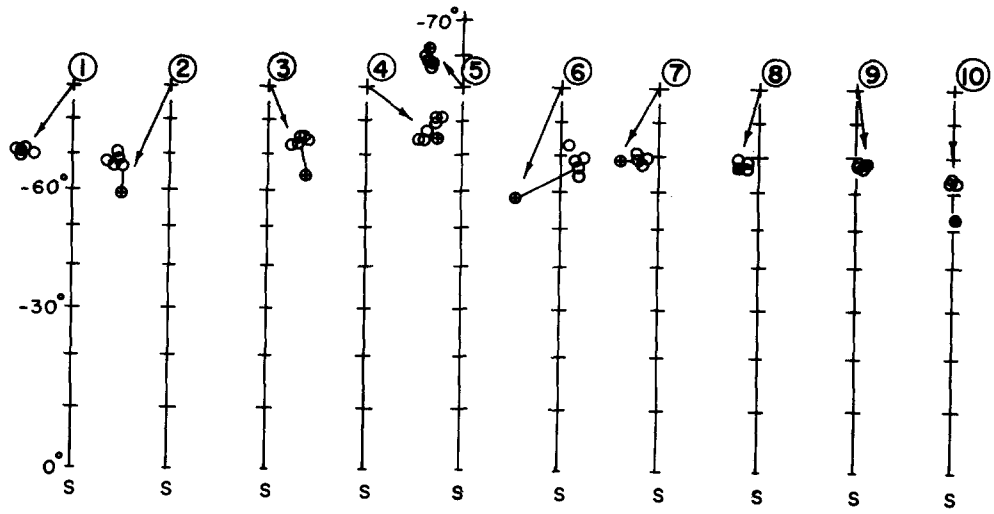


Fig. 3. Stereographic plots of directions of magnetization after successive a.f. cleaning steps. N.R.M. directions marked by crosses. Circled numbers refer to site numbers of pilot specimens.

All specimens were submitted to a magnetic field cleaning treatment in a peak demagnetizing field of 250 Oe and the cleaned magnetization directions were computed in the same way as the NRM directions. The angle θ between the two vectors obtained for 46 of the cores was used to appraise the magnetic homogeneity of the cores. Fig. 4 gives a histogram of this angle and illustrates clearly that the Lake Mistastin volcanics are very homogeneously magnetized in comparison with previously studied basic intrusive rocks [7-9]. This seems to justify basing the direction of magnetization of the 30 other cores on one observation per core.

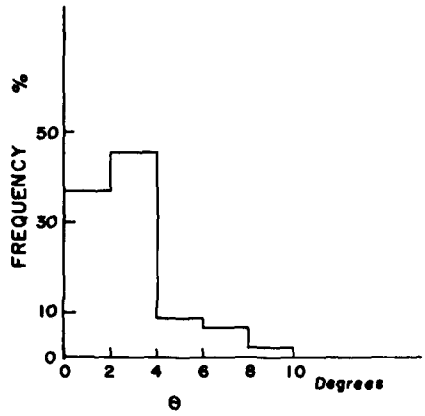


Fig. 4. Histogram of angle θ between pairs of magnetization directions obtained for each of 46 cores.

The directions of magnetization were grouped by sites and the mean direction for each site was computed vectorially. These site mean directions, and their standard errors, are listed in table 1. The standard errors of the site mean directions are defined by [10]

$$\delta_{\bar{R}_i} = \left[\frac{2(N_i - R_i)}{N_i(N_i - 1)}\right]^{\frac{1}{2}} = \frac{2}{\sqrt{kN_i}} \text{ radians},$$

where k is Fishers precision index. An examination of these results reveals that the magnetic cleaning treatment resulted in (1) increasing the inclination of the site means (away from the direction of the present field) without appreciably changing their declinations, and (2) decreasing significantly the standard error of the mean of site 3 without increasing that of any of the other site. The mean inclinations of sites 4, 5, and 7 are appreciably steeper than the other site means. After inspection of the data obtained for these sites it is evident that the abnormally steep inclinations are due to one abnormal direction in each case. The mean site direction and standard deviation δ'_R was re-computed for each of the 3 sites omitting the aberrant datum and the angle Ψ between the resultant (D', I') and the aberrant direction was determined. As shown in table 2 the angle Ψ is in each case more than 4 times as large as the standard deviation δ'_R.

Table 1
Site mean directions of magnetization.

Sample unit no.	Site no.	N	NRM				RM_{250} Oe			
			D (deg)	I (deg)	R	$\delta\bar{R}$ (deg)	D (deg)	I (deg)	R	$\delta\bar{R}$ (deg)
2	**1**	7	199.9	−62.4	6.904	3.9	201.0	−68.9	6.938	3.1
3	**2**	8	195.5	−54.5	7.751	5.4	186.7	−67.4	7.871	3.9
3	**3**	5	157.4	−54.4	4.613	11.3	177.7	−67.7	4.966	3.3
2	**4**	7	170.3	−68.7	6.215	11.1	157.2	−74.3	6.656	7.3
2	**5**	8	181.7	−71.4	7.796	4.9	179.0	−72.3	7.804	4.8
3	**6**	8	186.7	−53.3	7.885	3.7	170.8	−66.0	7.963	2.1
1	**7**	10	207.1	−72.5	8.979	8.6	204.4	−74.8	9.047	8.3
3	**8**	7	181.0	−67.9	6.976	1.9	179.2	−68.6	6.977	1.9
3	**9**	8	169.5	−65.6	7.751	5.4	169.9	−68.1	7.803	4.8
3	**10**	8	177.5	−60.5	7.952	2.4	174.6	−66.3	7.980	1.5

N = number of cores
D, I = declination and inclination of site mean.
R = modulus of resultant of N directions.
$\delta\bar{R}$ = standard error of site mean.

Note: in tables 1 and 2, data from the boldface numbers are used in the statistical analysis in table 3.

Table 2
"Adjusted" means of sites 4, 5 and 7.

Unit no.	Site no.	N'	D' (deg)	I' (deg)	R'	$\delta\bar{R}'$ (deg)	δ_R' (deg)	Ψ (deg)
2	4	6	165.1	−68.2	5.970	2.5	6.2	52
2	5	7	174.9	−68.6	6.946	2.9	7.7	37
1	7	9	196.7	−68.9	8.937	2.4	7.2	86

N = number of cores.
D, I = declination and inclination of site mean.
R = modulus of resultant of N directions.
$\delta\bar{R}$ = standard error of site mean.
δ_R' = standard deviation of N' directions = $\sqrt{N'}\delta\bar{R}'$.

According to our interpretation the probability that the abnormal direction belongs to the same population as the other directions from the site is lower than 0.001. Whether the 3 erratic directions result from major orientation errors or from drilling the cores from frost-heaved blocks cannot easily be determined. These directions cannot be related to likely variations in the direction of the paleomagnetic field. Therefore the site means listed in table 2 were used in the rest of this analysis, in preference to their counterparts in table 1.

6. STATISTICAL ANALYSIS

The data dealt with in the following analysis are summarized in tables 1 and 2. As a first step the 73 directions of magnetization considered reliable were

Table 3
Variance-ratio tests.

Units	N	B	D (deg)	I (deg)	δ (deg)	δ_b^2 rad^2	δ_w^2 rad^2	δ_b^2/δ_w^2	$F_{0.05}$	δ_m (deg)	δ_n (deg)
3 units	73	10	179.9	−68.2	8.8	0.04526	0.02056	2.20	1.68	4.4	4.4
No. 2	20	3	180.8	−69.3	9.0	0.09089	0.01720	5.28	2.65	6.7	6.7
No. 3	44	6	176.3	−67.4	8.5	0.01603	0.02309	0.69	1.96	2.6	2.6
3 units	(10)	(3)	179.4	−68.2	4.4	0.00829	0.00535	1.54	3.11	2.8	4.1

D, I = declination and inclination of mean of N vectors, i.e., of mean of B "weighed" site (unit) means.
δ = angular standard deviation of N vectors.
δ_b^2, δ_w^2 = between-site (unit) and within-site (unit) angular variances [10].
$F_{0.05}$ = theoretical statistic $F_{2(B-1),\,2(N-B),0.05}$.
δ_m = angular standard deviation of the B "weighed" site (unit) means = $\delta_b\sqrt{B/N}$.
δ_n = angular standard deviation of the B "normalized" site (unit) means = $[2(B-R)/(B-1)]^{\frac{1}{2}}$.

grouped by sites and a variance-ratio test [10] was used to determine on a statistical basis whether the site means differed significantly. If the ratio $\delta_b^2/\delta_w^2 > F_{0.05}$, the site directions are distinct, at the 95 percent confidence level. The result (1st line, table 3) yielded by the test suggests either that the age of the rocks varies from site to site or that they have been differentially disturbed since emplacement. Repeating the test for the directions of units 2 and 3 separately yielded an inconclusive answer since the site means were found significantly different within unit 2 but not within unit 3 (lines 2 and 3, table 3). Finally the normalized site means were grouped by units and the unit mean directions were computed. A variance-ratio test indicates that the unit means are not significantly distinct (line 4, table 3). The units may therefore be considered as paleomagnetically contemporaneous. The 10 site means may be used to estimate the position of the magnetic North Pole during a short period of upper Triassic time. The dispersion of the site means provides an estimate of the minimum angular standard deviation to be expected in a set of paleomagnetic directions representative of an area of about 100 km^2. Their standard deviation ($\delta_m = \delta_b(\sqrt{10/73})$ radians = 4.4 degrees) is clearly much less than what it would be if the Lake Mistastin volcanics had formed over a period of 10^4 years. It could reflect solely the effect of minor disturbances across the sampled area, to which should be added the effect of sampling and measurement errors. The limit of confidence about the resultant of the 10 site means may be defined either by its standard error ($\delta_{\overline{m}} = \delta_m/\sqrt{10}$) which is 1.4° in this case, or by the radius of the circle of confidence ($\alpha'_{0.95} = \sqrt{3}\,\delta_{\overline{m}}$ at the 0.95 probability level) 2.4° in the present case.

7. POLE POSITION

The only type of pole position derivable from the above data is a virtual pole in the sense defined by Doell and Cox [11]. This pole (117.7°E, 85.5°N) is in strikingly good agreement with the only reliable Jurassic pole (124.5°E, 85.5°N) obtained so far from North American rocks [12]. This coincidence suggests that the true age of the Lake Mistastin volcanics may be slightly younger than the single radiometric age obtained. If so, it would validate the assumption of a geocentric axial dipole field during the Jurassic. Both poles differ significantly from presently accepted Triassic or Cretaceous, North American poles [8]. In the light of these recent results, the use of paleomagnetism to determine the approximate age of Mesozoic rocks appears more promising than ever before.

8. SUMMARY AND CONCLUSION

The petrographic and magnetic homogeneity of the Lake Mistastin volcanics suggests that both the magnetite and the hematite in them are either primary

or deuteric minerals. The unaltered state of the rocks and their magnetic stability and homogeneity establish that their remanent magnetization is of thermoremanent origin. The high coherency of the data support the field observation that the 3 identified rock units are probably contemporaneous and practically undisturbed since emplacement. The virtual pole position derived from these data is in remarkable proximity to the Jurassic pole reported by Opdyke and Wensink for the White Mountain magma series of New Hampshire. This agreement suggests that the position of the pole in late Triassic and Jurassic time was significantly different from what it was in Cretaceous and early Triassic times. Finally, these data suggest that the paleomagnetic field in Jurassic time probably had a geocentric dipolar configuration.

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